

Kafka

1) Kafka Installation & Setup (Separate)

Aim:

To install and configure Apache Kafka.

Install Java

```
sudo apt update  
sudo apt install default-jdk-y  
java-version
```

Install SSH

```
sudo apt install openssh-server-y  
sudo service ssh restart
```

Download Kafka

```
wget https://archive.apache.org/dist/kafka/3.2.1/kafka_2.12-3.2.1.tgz
```

Extract Kafka

```
tar-xzf kafka_2.12-3.2.1.tgz
```

Install Zookeeper

```
sudo apt install zookeeperd-y
```

Start Zookeeper

```
bin/zookeeper-server-start.sh config/zookeeper.properties
```

Start Kafka Server

```
bin/kafka-server-start.sh config/server.properties
```

Result:

Kafka installation and setup completed successfully.

2) Kafka Streaming Application (Separate)

Aim:

To implement streaming using Apache Kafka.

Create Topic

```
bin/kafka-topics.sh --create \  
--topic test-topic \  
--bootstrap-server localhost:9092 \  
--partitions 1 \  
--replication-factor 1
```

Producer

```
bin/kafka-console-producer.sh \  
--topic test-topic \  
--bootstrap-server localhost:9092
```

Enter:

```
Hello  
Kafka  
Streaming
```

Consumer

(Open new terminal)

```
bin/kafka-console-consumer.sh \  
--topic test-topic \  
--from-beginning \  
--bootstrap-server localhost:9092
```

Output

```
Hello  
Kafka  
Streaming
```

Result:

Kafka producer-consumer streaming implemented successfully.